Mitutoyo Digimatic ↔ Arduino: Pinout & Wiring One■Pager

Safety First

Mitutoyo SPC lines are open■collector and low■voltage. Always use level shifting or optocouplers. Do NOT connect SPC lines directly to 5V Arduino pins.

Caliper Signal	Typical SPC Label	Arduino (via level shift)	Notes
Clock	CLK	D2 (INT0)	Interrupt-driven read on falling edge
Data	DATA	D4	Sampled on CLK falling edges (LSB-first)
Request	REQ (/REQ)	D3 (OUTPUT, active LOW)	Hold LOW ~120 ms to request a frame
Ground	GND	GND	Common ground between caliper interface

Frame Layout (Common Mapping)

52 bits total = 13 nibbles (4 bits each), transmitted LSB-first on DATA with CLK pulses. N0..N7 = BCD digits (least significant first). N8 = sign (bit3=1 negative). N9 = decimal places. N10 = units (0=mm, 1=inch). N11..N12 = status.

```
| N12 | N11 | N10 | N9 | N8 | N7 | N6 | N5 | N4 | N3 | N2 | N1 | N0 |
|---- |---- |---- |--- |--- |--- |--- |--- |--- |--- |--- |--- |--- |
   ^MSB                                                          LSB^
```

Usage Notes

• Some models free■run; set USE_REQ=false in code to just listen. • If digits look scrambled, try switching to RISING edge or invert DATA in software. • Use the serial print of raw nibbles to verify mapping on your device.